

NVIDIA Corp.

NVDA GTC Takeaways and Key Debates

We attended GTC the last two days and highlight our key takeaways below.

Key Debates

GB200 interconnect, how's it done?: Comments from the keynote led to conversation around how the new GB200 would be architected from an interconnect perspective. We would note that DGX Grace-Hopper systems didn't use Optical or AECs within the rack, so there is no material change vs. what NVDA is laying out in DGX Grace-Blackwell. What is changing though is the number of in-rack electrical connections, which is materially increasing (company noted 5k connections per rack) and should benefit MTSI (linear equalizer ACC content). Jensen repeatedly pointed out that a majority of the in-rack connections today are over NVLink and there is no opportunity for that to go Optical or to AEC in the Blackwell architecture. MRVL should continue to do the aggregation layer (Jensen at the morning Q&A said he "loved" optical). We still see total volumes of the GB200 as a minority of deployments vs. ODM solutions based on the reference architecture.

What's going on with H200/Blackwell pricing?: We have heard a lot of feedback from investors about pricing potentially coming in flatter both for the H200 as well as the Blackwell series. Jensen did make the comment in the morning session that he wants to "deliver extraordinary value for many, and extremely good value for all" and this should be read that the company is not raising pricing as materially as the company could. The company repeated says they price on value for the customer, so we do think pricing increases but clearly not as much as the 4x to 30x (training and inference, respectively) that the company outlined in terms of performance benefit, and pricing likely comes in at \$30K-\$40K per GPU. There will be material increases in both die cost (2 vs. 1) and also memory cost (192GB vs. prior 80GB).

Keynote Takeaways

The first portion of the event was focused on the Blackwell product cycle. NVDA officially announced B100, B200, and variations of Grace Blackwell CPU-GPU product. Blackwell will be a 2 die chip with an overall larger footprint than prior gen Hopper. It will have 8 HBM3e memory stacks and a total on-chip memory capacity of 192GB, on-par with MI300x. Despite significant gen/gen performance gains seen across all metrics, the company focused mainly on the uplift in inference workloads, where they expect Blackwell to deliver up to 30x the performance of Hopper in certain applications. Power consumption was another area of focus, as NVDA highlighted Blackwell's ability to train LLMs at roughly 25% of the power consumption of Hopper over the same training time frame. Other than new GPU architecture, the company added new hardware to the lineup via new NVLink products, which are discussed in further detail below. Outside of hardware, NVDA introduced NIMs (Nvidia Inference Microservices),

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NVDA	OVERWEIGHT
U.S. Semiconductors & Semiconductor Capital Equipment	NEUTRAL
Price Target	USD 850.00
Price (18-Mar-24)	USD 884.55
Potential Upside/Downside	-3.9%

U.S. Semiconductors & Semiconductor Capital Equipment

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which are software packages for custom and pre-trained models, mainly targeted at the enterprise market. This is part of a broader strategy to make the company an “AI foundry” and aid the enterprise ecosystem in deploying generative AI applications based around their own proprietary data. In our eyes, the last major point of emphasis was AI in the physical world via Robotics. Robotic hardware will be trained in NVDA Omniverse using digital twins to respond to real world scenarios with the first applications targeting the Industrial market. The event was, as expected, tech focused and showcased the many areas for AI ecosystem expansion.

Hello, Blackwell: NVDA officially revealed their Blackwell GPU architecture and announced the new additions to their AI accelerator lineup: B100, B200, Grace Blackwell (GB200), and branded systems based around the GPUs. Blackwell will be the first NVDA GPU architecture to feature a chiplet-based compute architecture, with each GPU comprising of 2 dies compute die tightly coupled and linked through a 10TB/s NV-HBI (Nvidia High Band Interface). This allows the two chiplets to function as a unified GPU within the packaging, but provide significantly more compute power gen/gen. The compute tiles contain 104B transistors each, which is over 2x greater than Hopper’s 80B, and will be fabbed on TSMC’s 4NP process. Memory capacity is substantially increased gen/gen with Blackwell composing of 192GB of HBM3e (8 stacks x 24G, 4 stacks per die) offering 8TB/s of memory bandwidth compared to Hopper’s 80GB of HBM3 with ~3TB/s bandwidth. Notably, HGX B100/B200 is rip/replace compatible with HGX H100 systems, allowing for an easy upgrade path and will retain the common 8xGPU/2xCPU configuration we saw with HGX H100. When paired with the Grace CPU and new networking products, Blackwell GPUs are expected to unlock additional system benefits via increased performance across the back-end. The company introduced new products to the NVLink family, namely 5th gen NVLink, a new NVLink Switch chip, and NVL72. Each Blackwell GPU is expected to support 18 5th gen NVLink connections, vs 1 NVLink connection per H100, and can support 50GB/s of bidirectional bandwidth. NVL72 is a liquid cooled in-rack spine switch responsible for networking 72 Blackwell GPUs within the DGX GB200. The company’s system architecture calls for 9 trays of switch chips within the rack that are expected to drive over 14TB/s of total system bandwidth. Overall, the company remains focused on the AI system-level opportunity and continues to drive performance gains both at the GPU level and system level.

NIMs, Lowering the Barriers to AI Adoption: NVDA plans to support the future of enterprise AI with its NIMs (NVDA Inference Microservices) offering. The offering will allow companies to access both custom models and pre-trained models, which can be purchased directly from NVDA, and easily deploy them as microservices either within the company’s own infrastructure or via the cloud. We believe this strategy is important for several reasons. First, the enterprise world has largely lagged the hyperscalers in terms of deploying and monetizing AI at scale. Giving large enterprises an easy way to deploy generative AI and explore high ROI applications is both good for the ecosystem and accretive to NVDA’s overall hardware business. The company has recently achieved record-high gross margins, which we believe have largely been a function of strong pricing power on hardware. Adding meaningful services revenue to the overall business model could provide another leg up in terms of corporate margin and provide stickier, higher visibility revenue streams going forward. While this initiative isn’t exactly new, and we recall the company attempting something similar with last year’s announcement of CloudDGX, we still see the overall initiative as a step in the right direction and shows their support for the broader AI ecosystem if nothing else.

Robotics the Next Frontier of AI: NVDA showcased their ability to bring AI to the physical world through demonstrations with companies such as Siemens, where they use the NVDA Omniverse to train Robots for factory automation use cases. The application of AI in Robotics is expected to deliver value to modern workplaces and drive better efficiency in manufacturing, healthcare, and automotives as early adopters. NVDA showcased AGX, the world’s first processor designed specifically for robotics with low power utilization and sensor processors for AI. The product

offering will pair AGX with the company's OVX Omniverse digital twin platform, hosted on MSFT Azure, where the robots are trained in the digital world to respond to various real-life situations.

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NVIDIA Corp. (NVDA, 18-Mar-2024, USD 884.55), Overweight/Neutral, CD/CE/J/K/M

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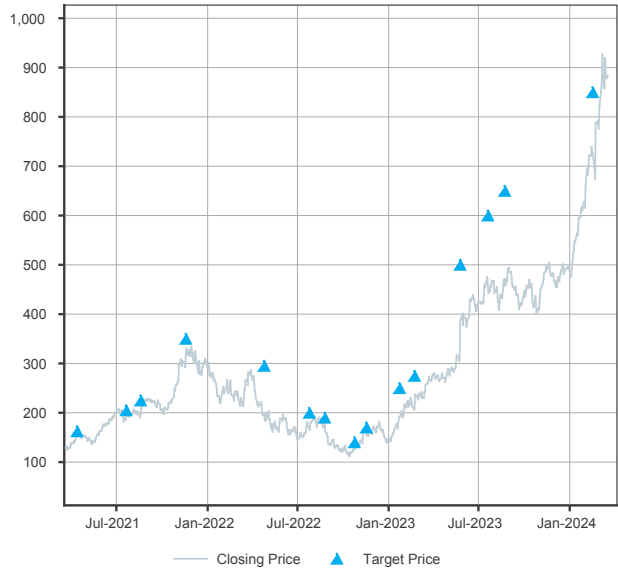
Stock Rating: **OVERWEIGHT**

Industry View: **NEUTRAL**

Closing Price: **USD 884.55** (18-Mar-2024)

Rating and Price Target Chart - USD (as of 18-Mar-2024)

Currency=USD



Source: IDC, Barclays Research

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Publication Date	Closing Price*	Rating	Adjusted Price Target
16-Feb-2024	726.58		850.00
23-Aug-2023	471.16		650.00
20-Jul-2023	470.77		600.00
25-May-2023	305.38		500.00
22-Feb-2023	207.54		275.00
23-Jan-2023	178.39		250.00
17-Nov-2022	159.10		170.00
24-Oct-2022	124.66		140.00
25-Aug-2022	172.22		190.00
25-Jul-2022	173.19		200.00
25-Apr-2022	195.15		295.00
18-Nov-2021	292.61		350.00
19-Aug-2021	197.98		225.00
21-Jul-2021	194.10		205.00
13-Apr-2021	156.80		162.50

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